

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

192511445

I SHAIK KHATIJAH RASHAN (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ShaiKh

Understanding of concept	Explanation	Application	Organisation	Accuracy	Clarity
25%	25%	20%	15%	10%	5%
5	5	4	3	2	1

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of ¹²6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits ⁶⁴~~48~~ frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted,

and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

C01-AT-1 $\Rightarrow$ SHORT ANSWER TEST

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COURSE:- CSA0702

COURSE NAME:- COMPUTER NETWORKS

SLOT $\rightarrow$ B

Question 1:- Answer:-

Given:- File size = 150 MB

1MB = 1,000,000 bytes.

Total File Size = $150 \times 1,000,000 = 150,000,000$ bytes.

Segment Size = 1500 bytes

Data link Layer Overhead = 40 bytes / frame

Band width = 50Mbps = 50,000,000 bits / second.

Calculating the no. of segments:

$$\begin{aligned} \text{No. of Segments} &= \frac{\text{File Size}}{\text{segment Size}} = \frac{150,000,000}{1500} \\ &= \underline{100,000} \text{ segments} \end{aligned}$$

calculating the no. of Frames:

each segment is placed into one frame.

Therefore, Number of Frames = 100,000

calculating total Data link layer Overhead.

Each frame adds 40 bytes.

Total overhead = $100,000 \times 40 = 4,000,000$ bytes.

Calculating Total Data Transmitted:

each frame contains:

* Payload = 1500 bytes , * Overhead = 40 bytes.

Total per frame:

$$1500 + 40 = 1540 \text{ bytes}$$

Total transmitted data:

$$100,000 \times 1540 = 154,000,000 \text{ bytes}$$

Convert to bits:

$$154,000,000 \times 8 = 1,232,000,000 \text{ bits}$$

Calculate Transmission Time:

$$\text{Transmission Time} = \frac{\text{Total Bits}}{\text{Bandwidth}} = \frac{1,232,000,000}{50,000,000} = 24.64 \text{ seconds}$$

$$\therefore \text{Estimated Transmission Time} = 24.64 \text{ seconds}$$

Question 2:- Answer:-

Given:-

- i) Sliding Window Size = 6 frames
- ii) Data per frame = 1024 bytes
- iii) Total Frames = 48 frames
- iv) Lost Frames = 1 frame in every window

Calculate the no. of Transmission windows:

$$\text{no. of windows} = \frac{\text{Total Frames}}{\text{Window Size}} = \frac{48}{6} = \underline{\underline{8 \text{ windows}}}$$

Calculate total Retransmissions:

One frame is lost in each transmission window

$$\therefore \text{Total Retransmissions} = 8 \times 1 = 8$$

$\therefore$ 8 retransmissions.

Total user Data Transmitted = $48 \times 1024 = \underline{49,152}$ bytes

Question 3 :- Answer:-

Given:-

i) Original IP Packet Size = 12,000 bytes

ii) MTU = 1500 bytes

iii) IP Header per Fragment = 20 bytes.

Calculate Maximum Data in Each Fragment

Maximum Data per Fragment = $1500 - 20 = 1480$ bytes

No. of Fragments = $\left\lceil \frac{\text{Original Packet Size}}{\text{Data per Fragment}} \right\rceil$

= $\left\lceil \frac{12000}{1480} \right\rceil = [8.108] = 9$ fragments

Total Header Overhead = $9 \times 20 = 180$ bytes.

First 8 fragments carry 1480 bytes of data each.

Remaining data :-

$12000 - (8 \times 1480) = 12000 - 11840 = 160$ bytes

Fragment 1-8 = 1480 bytes data + 20 byte header
= 1500 bytes

Fragment 9 = 160 bytes data + 20 bytes header
= 180 bytes.

$\therefore$

Maximum Data per Fragment = 1480 bytes
Total no. of Fragments = 9
Total Header Overhead = 180 bytes.

Question 4:- Answer:-

Given :-

i) Sliding window Size = 12 frames

ii) Data per Frame = 1024 bytes

iii) Total Frames = 64 frames

iv) lost Frames = 1 frame in every window.

$$\text{No. of windows} = \left\lceil \frac{\text{Total Frames}}{\text{Window Size}} \right\rceil = \left\lceil \frac{64}{12} \right\rceil = 5.33 = \underline{6} \text{ transmission windows}$$

$$\text{Total Retransmissions} = 6 \times 1 = 6 \text{ retransmissions.}$$

$$\begin{aligned} \text{Total user data Transmitted} &= 64 \times 1024 \\ &= 65,536 \text{ bytes} \end{aligned}$$

$\therefore$ Sliding Window Size = 12 frames

Total Frames = 64

Number of Transmission Windows = 6

Total Retransmissions = 6

Total User Data Transmitted = 65,536 bytes

Question :- 5 Answer :-

Given :-

i) Total File Size = 600 MB

ii) Synchronization Checkpoint = Every 60 MB

iii) Failure Occurs After = 390 MB

The no. of checkpoints before Failure :-

$$\begin{aligned} \text{No. of checkpoints} &= \left\lfloor \frac{\text{Data Transferred Before Failure}}{\text{Checkpoint Interval}} \right\rfloor \\ &= \frac{390}{60} = 6 \end{aligned}$$

check points are created at:

60 MB

120 MB

180 MB

240 MB

300 MB

360 MB

The next checkpoint (420 MB) is not reached because the

failure occurs at 390 MB.

$\therefore$ 6 check points.

Data that must be retransmitted = $390 - 360 = 30$ MB

$\therefore$ Total check points (created Before Failure) = 6

$\therefore$ Last Successful checkpoint = 360 MB

$\therefore$ Data to be Retransmitted = 30 MB

Question :- 6 :- Answer :-

Given:-

i) Original File Size = 900 MB

ii) Compression = 40%.

iii) Encryption Overhead = 5% of the compressed size.

$$\begin{aligned}\text{Compressed File size} &= \text{Original Size} \times (1 - 0.40) \\ &= 900 \times 0.60 \\ &= \underline{540 \text{ MB}}\end{aligned}$$

$$\text{Encryption Overhead} = 540 \times 0.05 = \underline{27 \text{ MB}}$$

$$\begin{aligned}\text{Encrypted File Size} &= 540 + 27 \\ &= \underline{567 \text{ MB}}\end{aligned}$$

$$\begin{aligned}\text{Percentage Reduction} &= \left(\frac{\text{Original Size} - \text{Final Encrypted Size}}{\text{Original Size}} \right) \times 100 \\ &= \frac{900 - 567}{900} \times 100 \\ &= \frac{333}{900} \times 100 = \underline{37\% \text{ reduction}}\end{aligned}$$

Original File Size = 900 MB

Compressed File Size = 540 MB

Encryption Overhead = 27 MB

Encryption File Size = 567 MB

Total Percentage Reduction = 37%

(5)



Question 7 :- Answer:-

Given :-

i) Total Data = 3GB

ii) 1GB = 1000MB

iii) Total Data = $3 \times 1000 = 3000 \text{ MB}$

iv) Request Size = 3MB

v) Throughput = 120 Mbps

$$\text{no. of requests} = \frac{\text{Total Data}}{\text{Request Size}} = \frac{3000}{3} = 1000 \text{ requests.}$$

$$\begin{aligned} \text{Convert Total Data into Bits} &= 3000 \text{ MB} \times 8 \\ \text{Here } 1 \text{ MB} &= 8 \text{ Mb} \end{aligned} = 24,000 \text{ Mb.}$$

$$\begin{aligned} \text{Transmission Time} &= \frac{\text{Total Data (Mb)}}{\text{Throughput (Mbps)}} \\ &= \frac{24,000}{120} = \underline{\underline{200 \text{ seconds.}}} \end{aligned}$$

Total Data = 3GB (3000 MB)

Request Size = 3MB

Total no. of Requests = 1000

Throughput = 120 Mbps.

Estimated Transmission Time = 200 seconds

Question :- 8:- Answer:-

Given:- Application Layer Delay = 4ms

Transport Layer Delay = 3ms

Network Layer Delay = 5ms

Data Link Layer Delay = 2ms

Physical Layer Delay = 1ms

Propagation Delay = 18ms

Transmission Delay = 12ms

$$\text{Total End-to-End Delay} = \text{Processing Delay} + \text{Propagation Delay} + \text{Transmission Delay}$$

$$= 15 + 18 + 12$$

$$= \underline{45 \text{ ms}}$$

$$\text{Total Processing Delay} = 15 \text{ ms}$$

$$\text{Propagation Delay} = 18 \text{ ms}$$

$$\text{Transmission Delay} = 12 \text{ ms}$$

$$\text{Total End to End Delay} = 45 \text{ ms}$$

Question 9:- Answer:-

Given:-

$$\text{Useful Data} = 80 \text{ MB}$$

$$1 \text{ MB} = 1,000,000 \text{ bytes}$$

$$\text{Payload} = \text{Frame Size} - \text{Overhead}$$

$$= 1600 - 80$$

$$= 1520 \text{ bytes}$$

$$\begin{aligned} \text{no. of Frames} &= \frac{\text{Useful Data}}{\text{Payload per Frame}} \\ &= \frac{80,000,000}{1520} \\ &= 52,632 \text{ frames} \end{aligned}$$

$$\begin{aligned} \text{Total Overhead} &= \text{No. of Frames} \times \text{Overhead per Frame} \\ &= 52,632 \times 80 \\ &= 4,210,560 \text{ bytes} \end{aligned}$$

$$\begin{aligned} \text{Transmission Efficiency} &= \frac{\text{Useful Data}}{\text{Useful Data} + \text{Overhead}} \times 100 \\ &= \frac{80,000,000}{80,000,000 + 4,210,560} \times 100 \\ &= 95\% \end{aligned}$$

Question 10: - Answer :-

Calculate the Compressed File Size by 30%.

$$\begin{aligned}\text{Compressed File Size} &= 240 \times (1 - 0.30) \\ &= 240 \times 0.70 = 168 \text{ MB}\end{aligned}$$

Convert compressed File Size into Bytes

$$168 \text{ MB} = 168 \times 1,000,000 = 168,000,000 \text{ bytes}$$

$$\text{no. of segments} = \frac{\text{compressed File Size}}{\text{Segment Size}} = \frac{168,000,000}{1400}$$

$$\begin{aligned}\text{Calculate Total Protocol overhead} &= 120,000 \times 60 \\ &= 7,200,000 \text{ bytes.} \\ &= 7.2 \text{ MB.}\end{aligned}$$

$$\begin{aligned}\text{Final Data} &= \text{Compressed Data} + \text{Protocol Overhead} \\ &= 168,000,000 + 7,200,000 = 175,200,000 \text{ bytes}\end{aligned}$$

Convert to MB:

$$175,200,000 \div 1,000,000 = 175.2 \text{ MB}$$

ASSESSMENT TOOL-1 (SHORT ANSWER TEST)

Course code	CSA0702	Course Title	Computer Networks
Co Assessed	CO1	Assessment	Tool- short Answer
Weightage	40%.	Total marks	20 marks

Student name	Shaik Khajah Roshan	Reg. No.	192511445
Section Batch	B.E. CSE (2025-2029)	Date of Submission	17, Jul 2026
Faculty Incharge	Dr. R.S. Selvan	Assessment	Individual Submission

Total Marks	20	Obtained Marks	20
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